Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/26/2006
PI Name:	Chang, Polly Y. Ph.D.		
Project Title:	Tissue-Specific Acute And Late Molecular Surveillance Of Particle Radiation Effects		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Radiation biology		
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	94025-3453	**Congressional District:**	18
Comments:			
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 01/31/2008 per A. Chu/ARC (7/07)		
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Task Description:	The detection of particle radiation induced alterations in the genome in different tissues in vivo is important for the assessment of both short- and long-term risks involved in space explorations. However, such studies are difficult due to (a) the lack of adequate selectable somatic target genes in mammalian systems and (b) the clonogenic potential of the somatic cells chosen. Mutagenic changes that affect the function of non-dividing tissues may be especially deleterious during long-term space flights. The advent of transgenic rodent gene mutation models has provided a means to rapidly assay, in a statistically reliable way, for mutations in the DNA from any tissue of the body over the period of the life span of the animal. These models are the only systems currently available for evaluating both short and long-term tissue specific mutagenic events. They have been accepted by regulatory agencies and are used in genetic toxicology studies on a wide variety of clastogenic and mutagenic agents. In the work proposed here, we have chosen to use a unique plasmid-based lacZ transgenic mouse system to determine the mutagenic potential of charged particle radiations relevant to the space environment. The lacZ transgenic mouse has been constructed so that every cell of the animal contains multiple copies of an integrated target gene. The use of this system allows us to recover and quantify radiation-induced mutations in all tissues of the animal, especially in tissues with limited cell turnover potential, and therefore allows tissue-specific radiation-induced changes to be measured as a function of time post radiation. The nature of these mutations can also be revealed by restriction fragment length polymorphisms (RFLP) and sequence analysis. We hypothesize that physical parameters of HZE radiation induces tissue-specific responses and aim to use the transgenic animal model in a multi-prong approach to extend our current knowledge to test three specific aims: (1) Examine the changes in transgene MF as a function of fluence, dose protraction of either pure or mixed particle beams. Beam characteristics in space are critical variables in space and tissue-specific biological effects are important in determining relative risks in space explorations; (2) Measure the dose- and temporal- dependence of HZE radiation-induced MF in the transgene in liver tissue and compare the results with those previously measured in the brain and the spleen. Excess relative risk in liver cancer and non-cancer liver disease or cirrhosis has been demonstrated in Atomic bomb survivors and thorotrast patients; (3) Advance our understanding of the molecular basis for mutation by surveying the transcription activation of target genes in the brain tissue. We will focus our research on examining the expression of proinflammatory cytokines, TNFalpha and interleukin-1, and the neuronal cell adhesion molecule NCAM-1. These genes have been shown to be induced in the mouse brain after low LET exposures. Knowledge gained in this ground-based work is directed at understanding tissue and organ response to space radiation and will aid both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate the negative effects in space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The space atmosphere is comprised of particle radiation in a wide spectrum of higly energetic and charged (HZE). Exposures to these HZE radiations poses a serious health hazard to humans in lng-term manned space explorations. We are using the plasmid-based lacZ transgenic mouse mutation model system to examine the acute and long-term tissue specific mutagenic potential after exposure to radiation with different qualities. Knowledge gained in this ground-based work is directed at understanding tissue and organ response to space radiation and will aid both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate the negative effects in space flights.
Task Progress:	In order to evaluate the impact of dose protraction on the mutagenic and genotoxic potential in vivo, lacZ transgenic mice were exposed to an acute or fractionated dose of high energy protons at the NSRL in the Fall of 2005. One cohort of animals was exposed to a single dose of 1 GeV/u protons while another cohort was exposed to 5 daily low dose (10 to 100 cGy) fractions, such that the cumulative dose for each dose group equals those from the acute exposures. Flow cytometric technique was used to map the changes in the frequency of MN-RET (fMN-RET) in peripheral blood samples in an early time course after exposure. Brain and spleen tissues were harvested at 2 to 4 months after exposure and mutant frequencies (MF) in the transgene in these tissues were measured. We noted a dose-dependent dramatic drop in the frequency of total circulating reticulocytes (fRET) shortly after an acute dose of proton irradiation, a definitive indicator that particle radiation induced an overall dose-dependent systemic toxicity in the hematopoietic system. fRET was depressed up to 72 hr after radiation exposure but rebounded with fRET exceeding control levels at 1 week after exposure. Radiation induced systemic genotoxicity was also immediately evidence by an increase in fMN-RET within 24 hr after irradiation. The dose-dependent elevation of fMN-RET peaked at 48 hr after exposure for doses less than or equal to 1 Gy. No significant elevated levels of fMN-RET, when compared to the sham-treated controls, were present at 1 week after exposure, suggesting that the cytogenetic damage in the hematopoietic system was transient. Animals were exposed to 5 daily low-dose (10 to 100 cGy) fractions of 1GeV/n protons to simulate protracted radiation exposures in the space environment. No significant changes in the fRET and fMN-RET were observed in the samples obtained from animals exposed to less than or equal to 2 Gy total proton dose at any time after fractionated exposures. Results from this study suggest that active and rapid recovery mechanisms from hematopoietic radiation damage are in place in this in vivo model system. Since the earliest damaged cells to enter the peripheral blood compartment are the cells that were in G2/M at the time of radiation exposure, differences in the fMN-RET at this early time may be used as an early assessment of biological effectiveness of particle beams in the hematopoietic stem cell population. We analyzed the lacZ mutant frequencies (MF) in the brain and spleen tissues harvested after exposure to 1 GeV/ n protons. Unlike the results obtained from 250 MeV/n protons, the MF in both tissues were minimally enhanced after 1 Gy of high energy protons. Fractionation did not result in an increase in transgene MF in either tissue at this dose. Further studies are needed to explore the significance of the initial energy of the proton beam on mutagenicity in tissues. We examined the lacZ MF in the liver tissue after 1 GeV/n iron ion irradiation. The spontaneous MF in the liver at 8 weeks was about 5.14 x 10E-5. These results are consistent with published values and within the expected range of liver MF for this model. 2 Gy iron ions irradiation induced a 1.5 fold increase in mutant frequencies in the liver. Work on evaluating the mutant characteristics from this tissue is in progress. To monitor for the expression of endogenous genes known to be important for normal maintenance of brain function, total RNA was extracted from whole brain. We quantitatively evaluated, using real-time PCR techniques, the transcription of tumor necrosis factor (TNFalpha), interleukin-1 (IL-1), and neural cell adhesion molecule-1 (NCAM-1) at 1 week after a range of 250 MeV/n proton doses. We observed a dose-dependent increase in the level of NCAM transcripts, reaching 2.5-fold higher than control levels after 2 Gy. The induction appears to be saturated with no further increases in expression at doses >2Gy. Similarly, TNFalpha expression also followed the same trend, also reaching >2.5-fold higher expression after 2 Gy when compared with the control. This saturation effect may be due to cell

	attrition as the high doses. Unlike TNFalpha and NCAM, IL1 expression peaked at about 2-fold higher than the control level at 1 Gy exposure, and declined at higher doses. These findings imply that different genes in the brain tissue are responsive to different doses of radiation. The pattern of NCAM expression after 1 GeV/n iron ions is distinctly different from those obtained after protons. We observed a persistent dose-dependent down-regulation of NCAM expression. NCAM expression in samples exposed at 2 Gy is about 30% of the control level. TNFalpha expression was also down-regulated in the brain tissue, with levels below control levels after 0.5, 1, or 2 Gy of Iron particles. These quantitative results suggest that gene expression in the brain tissue is highly dependent on the quality of particle radiation. Recent evidence suggests that a number of key features of the synapse are regulated by several families of cell-adhesion molecules (CAMs). The CAMs that are enriched at synaptic junctions include NCAM, L1-CAM, cadherins, protocadherins, and integrins. These molecules are known to be involved in contact initiation, recruitment of presynaptic and postsynaptic proteins, synapse maturation/stabilization or elimination, and synaptic plasticity in the mature brain. NCAMs are members of the immunoglobulin superfamily and are closely related to cell surface glycoproteins. Several studies report that NCAMs play a central role in memory. Our current findings uncovering the dysregulation of NCAM in brain tissue at 1 week after iron ion exposure implicates neuronal communication and signaling consequences that could contribute to degenerative brain tissue risks from space radiation. Results presented here validate our initial hypothesis that high LET exposure elicits an inflammatory response in the brain shortly after irradiation. Such early responses result in alterations in the coordination of normal signaling pathways that are important for the maintenance of normal brain function and cognition.
Bibliography Type:	Description: (Last Updated: 04/26/2023)
Abstracts for Journals and Proceedings	Chang PY. "Transgenic Mouse Mutation Models and Heavy Ion Effects." Invited speaker: Symposium on Ion Beams from Physics to Therapy, Radiation Research Society Annual Meeting, Denver CO, October 15th – 19th 2005. Proceedings of the Radiation Research Society Meeting, Oct 2005. , Oct-2005
Abstracts for Journals and Proceedings	Chang PY, Bakke J, Puey A, Lin S. "Particle-radiation induced cytogenetic and mutagenic effects in vivo." Presented at the 4th International Workshop on Space Radiation Research, Moscow/St. Petersburg, Russia, June 5 – 9, 2006. Proceedings, 4th International Workshop on Space Radiation Research, p. 20, June 2006. , Jun-2006
Articles in Peer-reviewed Journals	Chang PY, Bakke J, Orduna J, Lin S, Doppalaudi R. "Proton-induced genetic damage in lacZ transgenic mice." Radiat Res. 2005 Oct;164(4 Pt 2):481-6. PMID: 16187753 , Oct-2005